

Let $\langle, \rangle : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ be a mapping given by $(\vec{x}, \vec{y}) \mapsto \vec{x}^T \cdot A \cdot \vec{y}$ where $A_{2 \times 2}$ is a real symmetrical square matrix. Prove that $\langle, \rangle$ is a scalar multiplication mapping if and only if $a > 0$ and $\det A > 0$

$$A := \begin{pmatrix} a & b \\ b & c \end{pmatrix}$$

Then (SSS) is the result of the symmetry of A while (SL1) and (SL2) are the results of matrix multiplication. The last necessary proof is (SP) or that $\|\vec{x}\| > 0$ for all $\vec{x} \neq \vec{0}$ and equal to 0 only when $\vec{x} = \vec{0}$. Let's examine a case for any $\vec{x} \in \mathbb{R}^2$

$$\langle \vec{x}, \vec{x} \rangle = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} a & b \\ b & c \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = ax_1^2 + 2b(x_1x_2) + cx_2^2 =$$

This we can further manipulate until we get an expression containing squares of values x_1 and x_2

$$= a \left(x_1^2 + 2\frac{b}{a}(x_1x_2) + \left(\frac{b}{a}\right)^2 x_2^2 - \left(\frac{b}{a}\right)^2 x_2^2 + \frac{c}{a}x_2^2 \right) = a \left(x_1 + \frac{b}{a}x_2 \right)^2 + \frac{ac - b^2}{a}x_2^2$$

($\implies$) Assuming that $a > 0$ and $\det A > 0$ or $ac - b^2 > 0$ the expression is always positive except when $\vec{x} = \vec{0}$; which is the only case when it equals to 0.

($\impliedby$) Assuming that $\langle \vec{x}, \vec{x} \rangle > 0$ and $a \neq 0$ we can look at this expression in two cases

1. $a > 0$

The first square is always greater or equal to 0.

Assuming the second square has a negative coefficient we can plug in a non zero vector like $(-b/a \cdot x_2, x_2)^T$ and the expression collapses to $\frac{ac - b^2}{a}x_2^2$ which is less than 0. $\nexists$

Assuming the second square coefficient is 0, plugging in the same vector would give us 0 even for a $x_2 \neq 0$. $\nexists$

Therefore $ac - b^2 = \det A > 0$.

2. $a < 0$

We simply have to plug in a non zero vector like $(x_1, 0)^T$ and the expression collapses to ax_1^2 which is obviously less than 0. $\nexists$

Therefore a cannot be less than 0

Lastly what if $a = 0$. Then $ax_1^2 + 2b(x_1x_2)cx_2^2 = 2b(x_1x_2) + cx_2^2$. For any $b, c \in \mathbb{R}$ we can plug in a non zero vector $(x_1, 0)^T$ and the expression collapses to 0. $\nexists$

Therefore $a \neq 0$.

Q. E. D.